3.9 Mathematical Requirements

In order to develop their skills, knowledge and understanding in science, candidates need to have been taught, and to have acquired competence in, the appropriate areas of mathematics relevant to the subject as set out below.

	Candidates should be able to:
Arithmetic and computation	• recognise and use expressions in decimal and standard form • use ratios, fractions and percentages • make estimates of the results of calculations (without using a calculator) • use calculators to find and use power, exponential and logarithmic functions (x^n, $1/x$, $\sqrt{x}$, $\log_{10}x$, e^x, $\log_e x$)
Handling data	• use an appropriate number of significant figures • find arithmetic means • construct and interpret frequency tables and diagrams, bar charts and histograms
Algebra	• understand and use the symbols: $=$, $<$, $<<$, $>>$, $>$, $\propto$, $\sim$. • change the subject of an equation by manipulation of the terms, including positive, negative, integer and fractional indices • substitute numerical values into algebraic equations using appropriate units for physical quantities • solve simple algebraic equations • use logarithms in relation to quantities that range over several orders of magnitude
Graphs	• translate information between graphical, numerical and algebraic forms • plot two variables from experimental or other data • understand that $y = mx + c$ represents a linear relationship • determine the slope and intercept of a linear graph • calculate rate of change from a graph showing a linear relationship • draw and use the slope of a tangent to a curve as a measure of rate of change
Geometry and trigonometry	• appreciate angles and shapes in regular 2D and 3D structures • visualise and represent 2D and 3D forms including two-dimensional representations of 3D objects • understand the symmetry of 2D and 3D shapes